

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION**Revision Test 2026****Chemistry — Class 9 (Science Group)**

Syllabus: Chapter 1 — Nature of Chemistry | Chapter 2 — Matter | Chapter 3 — Atomic Structure | Chapter 4 — Periodic Table & Periodicity | Chapter 5 — Chemical Bonding | Chapter 6 — Stoichiometry | Chapter 7 — Electrochemistry | Chapter 8 — Energetics | Chapter 9 — Chemical Equilibrium

Total Marks: 43**Time: 2 Hours****Paper: Exam-19 (Version 1)**

1. Attempt ALL questions. There are no choices in this paper.
2. Section A (24 marks): 24 MCQs. Fill the correct bubble. Suggested time: 30 minutes.
3. Show complete working for every calculation. Always write the correct unit with your final answer.
4. Use black or blue ink only. Pencil is allowed only for diagrams. Relative atomic masses are given where needed.

SECTION A — Multiple Choice Questions (24 × 1 = 24 Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	Chemistry is best defined as the study of:	living organisms and their life processes	matter, its properties, composition and its interactions with other matter and energy	the motion of bodies and the forces acting on them	the structure of the Earth and its rocks	(A)(B)(C)(D)
2	Which state of matter can be compressed most easily?	solid	liquid	gas	all three equally	(A)(B)(C)(D)
3	Which one of the following is a compound?	air	brass	carbon dioxide	sea water	(A)(B)(C)(D)
4	A solution that can dissolve no more solute at a given temperature is said to be:	saturated	unsaturated	dilute	colloidal	(A)(B)(C)(D)
5	Inside the nucleus of an atom, the protons and neutrons are held together by the:	electrostatic force	gravitational force	magnetic force	strong nuclear force	(A)(B)(C)(D)
6	As the shell number of an electron increases, its energy and its average distance from the nucleus:	both decrease	both increase	energy increases, distance decreases	energy decreases, distance increases	(A)(B)(C)(D)
7	The nucleon number (mass number) of an atom is the sum of the number of:	protons and neutrons	protons and electrons	neutrons and electrons	protons only	(A)(B)(C)(D)
8	Two isotopes of the same element have:	different numbers of protons	the same number of neutrons	the same chemical properties but different molecular masses	completely different chemical properties	(A)(B)(C)(D)
9	In the modern periodic table, the elements are arranged in order of increasing:	atomic mass	proton number	number of neutrons	density	(A)(B)(C)(D)
10	An element belonging to Group 2 of the periodic table forms an ion of charge:	−1	−2	+1	+2	(A)(B)(C)(D)
11	On moving down a group in the periodic table, the atomic radius of the elements:	increases	decreases	remains the same	first decreases,	(A)(B)(C)(D)

					then increases	
12	The elements of Group 17 of the periodic table are collectively known as the:	alkali metals	alkaline earth metals	halogens	noble gases	(A)(B)(C)(D)
13	A metallic bond is the strong electrostatic attraction between:	two oppositely charged ions	a sea of delocalized electrons and positively charged metal cations	a shared pair of electrons and two nuclei	two neutral atoms of a metal	(A)(B)(C)(D)
14	Metals are described as electropositive elements because they tend to:	gain electrons	share electrons equally	attract electrons strongly	lose electrons	(A)(B)(C)(D)
15	Compared with covalent compounds, ionic compounds generally have:	higher melting points	lower melting points	no melting point at all	exactly the same melting points	(A)(B)(C)(D)
16	Atoms of main group elements take part in bonding in order to achieve the electronic configuration of the nearest:	alkali metal	halogen	noble gas	transition element	(A)(B)(C)(D)
17	The empirical formula of a compound gives:	the actual number of atoms in one molecule	the mass of one molecule of the compound	the number of moles in one gram	the simplest whole number ratio of the different atoms	(A)(B)(C)(D)
18	The number of moles present in 80 g of sodium hydroxide, NaOH, is: (Na = 23, O = 16, H = 1)	1 mol	2 mol	4 mol	0.5 mol	(A)(B)(C)(D)
19	In a chemical equation, the state symbol (aq) shows that the substance is:	dissolved in water	a gas	a pure liquid	a solid	(A)(B)(C)(D)
20	The formula of the compound formed between calcium ions (Ca^{2+}) and nitrate ions (NO_3^-) is:	CaNO_3	Ca_2NO_3	$\text{Ca}(\text{NO}_3)_3$	$\text{Ca}(\text{NO}_3)_2$	(A)(B)(C)(D)
21	When the equation $__\text{Zn} + __\text{O}_2 \rightarrow __\text{ZnO}$ is correctly balanced, the coefficients are:	1, 1, 1	2, 1, 2	1, 2, 2	2, 2, 1	(A)(B)(C)(D)
22	The oxidation number of an element in its free (uncombined) state is:	+1	-1	zero	equal to its group number	(A)(B)(C)(D)
23	The oxidation number of a monoatomic ion is:	the same as the charge on the ion	always zero	always +1	equal to its atomic number	(A)(B)(C)(D)
24	Activation energy is defined as the minimum energy that colliding molecules must have for a:	reversible reaction	change of physical state	catalysed pathway	successful collision	(A)(B)(C)(D)

SECTION B — Short Questions (3 × 3 = 9 Marks)

Q.2	Attempt all three parts.	Marks
(i)	Calcium (Z = 20) reacts with fluorine (Z = 9) to form calcium fluoride, CaF_2 . (a) Show the formation of CaF_2 by drawing a dot-and-cross diagram, showing the electron transfer clearly. [2] (b) Name the type of bond formed and define it. [1]	3
(ii)	Lithium occurs naturally as two isotopes: ^6Li with an abundance of 7.5% and ^7Li	3

	with an abundance of 92.5%. (a) Calculate the relative atomic mass of lithium, showing your working. [2] (b) State the number of protons, neutrons and electrons in one atom of ${}^7\text{Li}$. [1]	
(iii)	An atom of beryllium has atomic number $Z = 4$. (a) Write its complete electronic configuration. [1] (b) State its group and its period in the periodic table, giving a reason for each. [1] (c) State the charge on the ion that beryllium forms, and explain how its group number tells you this. [1]	3

SECTION C — Long Questions (2 × 5 = 10 Marks)

Q.3	Attempt both questions.	Marks
(i)	(a) Define the terms <i>empirical formula</i> and <i>molecular formula</i> . [1] (b) On analysis, a white solid was found to contain 56.5% potassium, 8.7% carbon and 34.8% oxygen by mass. Its relative formula mass is 138. Determine its empirical formula and its molecular formula, showing every step of your working. [3] ($K = 39$, $C = 12$, $O = 16$) (c) This solid reacts with dilute hydrochloric acid. Balance the following equation and add the correct state symbols to every substance: [1] $\text{___ K}_2\text{CO}_3 + \text{___ HCl} \rightarrow \text{___ KCl} + \text{___ H}_2\text{O} + \text{___ CO}_2$	5
(ii)	A sample of magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$, has a mass of 22.2 g. ($Mg = 24$, $N = 14$, $O = 16$, $N_A = 6.022 \times 10^{23}$) (a) Calculate the molar mass of magnesium nitrate. [1] (b) Calculate the number of moles of magnesium nitrate present in the sample. [2] (c) Calculate the number of formula units of magnesium nitrate present in the sample. [1] (d) State the oxidation number of magnesium in this compound, and give the rule that tells you its value. [1]	5

Student's Signature: _____

Invigilator: _____

Date: _____

— End of Question Paper —